



DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF PHYSICAL SCIENCES
UNIVERSITY OF BENIN, BENIN CITY

2024/2025 BSC COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS
COURSE: COS201 - COMPUTER PROGRAMMING I

TIME: 1hr

Instruction: Shade the correct option on your OMR Answer sheet

1. _____ is not a feature of Java?
(a) Object-Oriented (b) Platform Dependent
(c) Secure (d) Robust (e) None of the above

2. Java does not support _____
(a) Multiple inheritance through classes (b) Object-oriented programming (c) Interface implementation (d) Exception handling (e) All of the above

3. How many primitive data types are there in Java?
(a) 6 (b) 7 (c) 8 (d) 9 (e) 10

4. What will be the output of the following code?

```
int x = 5;
int y = x;
x = 10;
System.out.println(y);
```

- (a) 10 (b) 5 (c) 0 (d) Compilation Error
(e) Syntax Error

5. Which of the following statements about reference types is true? (a) Reference types store values directly (b) Reference types are not objects (c) Reference types store memory addresses (references) (d) Reference types can't be null (e) None of the above

6. Which of the following statements is true regarding procedural vs object-oriented paradigms in Java?
(a) Java does not allow procedural programming because all methods must be non-static.
(b) Procedural programming is discouraged but still syntactically valid in Java.
(c) Procedural programming requires inheritance to implement in Java.
(d) Procedural and object-oriented programming are mutually exclusive in Java (e) None of the above.

7. What procedural programming characteristic is demonstrated by this code?

```
public class Calculator {
    public static int sum(int a, int b) {
        return a + b;
    }
    public static void main(String[] args) {
        int result = sum(10, 20);
        System.out.println(result);
    }
}
```

- (a) Data encapsulation (b) Inheritance and polymorphism (c) Use of static functions without objects (d) Dynamic method dispatch (e) Functional programming

8. _____ does not align with Java feature of being a robust language. (a) Automatic garbage collection (b) Strong memory management (c) Lack of pointer arithmetic (d) Extensive use of goto statements (e) Multithreading

9. Java programs are compiled into an intermediate form called _____. (a) Machine code (b) Source code (c) Bytecode (d) Assembly code (e) Tinycode

Using the code snippet below, answer questions 10 – 11

```
import java.util.Scanner;
public class UserInputExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = scanner.nextLine();
        System.out.println("Hello, " + name);
        scanner.close();
    }
}
```

10. Why is scanner.nextLine() used instead of scanner.next() in this program? (a) nextLine() ignores user input (b) nextLine() reads only the

first word (c) next line) reads the entire line including spaces (d) next line() is deprecated. (e) Don't Know why

11. Which package must be imported to use the Scanner class in Java? (a) java.io (b) java.util (c) java.lang (d) java.input (e) java.string

12. What is variable shadowing in Java? (a) When two methods have the same name but different parameters (b) When a subclass defines a field with the same name as in its superclass (c) When a constructor calls another constructor in the same class (d) When a local variable is declared inside a loop (e) None of the above

13. Which cast is allowed without an explicit cast operator? (a) int to long (b) long to int (c) Object to String (d) Double to float (e) object to int

14. What is method overloading in Java? (a) Defining multiple methods with the same name and same parameters (b) Defining methods with different names (c) Defining multiple methods with the same name but different parameters (d) Using one method in another class (e) None of the above

Consider the code snippet below and answer questions 15 to 17

```
1 public class Dog {
2     int Height;
3     private int weight;
4
5     public Dog() {
6         weight = 42;
7     }
8
9     public void cat(){
10
11     }
12
13 }
```

15. How many method(s) does the class Dog contain? (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

16. What is the Access Specifier of the variable Height? (a) Public (b) Private (c) Protected (d) None of the Above (e) All of the Above

17. Which of the following is true of the class Dog?

- (i) The name of the file is Dog.java
(ii) The name of the file is unknown
(iii) The class does not have a default constructor
(iv) The variable weight is encapsulated
(a) I Only (b) I and II (c) III (d) III and IV (e) IV only

18. The code below will not run because of an error. What should be done to resolve the issue?

```
int a = 40;
short b = a;
```

- (a) Garbage Collection (b) Casting (c) Overloading (d) Polymorphism (e) Instantiation

Consider the code below and answer questions 19 to 21:

```
public class Student{
    String Name;
    private int Age;
    Student John;
    void setAge(int Age){
        this.Age=Age;
    }

    char Sex;
    int getAge(){
        Return Age;
    }

    public void createObject(){
        John = new Student();
    }
}
```

19. What type of variable is the variable, John? (a) Local Variable (b) Automatic Variable (c) Instance variable (d) Static Variable (e) None of the Above

20. Which of the following variables is an instance Variable? (a) Name (b) Age (c) Sex (d) All of the Above (e) None of the Above

21. What is the return type of the setAge method? (a) int (b) char (c) String (d) no return type (e) default

22. Identify the return type of a method that does not return any value. (a) int (b) void (c) double (d) return (e) None of the Above

23. What are the variables declared in a class for the use of all methods of the class called? (a) Local Variables (b) Instance variables (c) Reference variable (d) Class variables (e) None of the Above

24. Identify the prototype of the default constructor in the following code:

```
public class Solution {
```

(a) Solution(void) (b) void Solution() (c) public Solution(void) (d) public Solution() (e) None of the Above

25. Which one of the following is not an access specifier? (a) Protected (b) Static (c) Public (d) Private (e) None of the Above

26. Which of the following keywords allows a class to achieving multiple inheritance? (a) extends (b) implements (c) inherits (d) uses (e) None of the Above

27. How can you achieve polymorphism in Java? (a) By method overloading (b) By method overriding (c) By both method overloading and overriding (d) By extending classes (e) None of the Above

28. What happens if a method in a subclass has the same name and parameters as a method in its superclass? (a) The superclass method is executed (b) The subclass method is executed (c) A compile-time error occurs (d) The program fails to run (e) The subclass method is overridden

29. Which of the following is valid for single inheritance?
(a) class SubClass extends SuperClass {}
(b) Subclass inherits SuperClass {}
(c) class SubClass implements SuperClass {}
(d) SubClass : SuperClass {}
(e) Subclass extends SuperClass {}

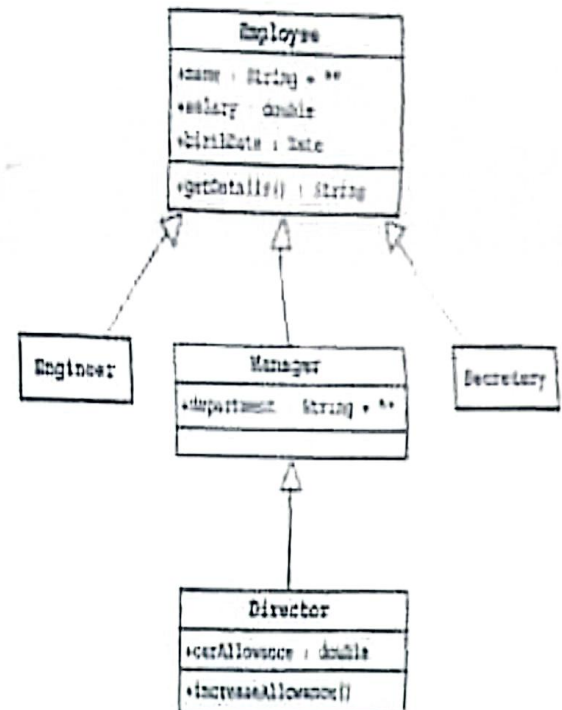
30. Which operator is used to instantiate a class? (a) extends (b) new (c) constructor (d) plus (e) reference

31. Find the output of the following program.

```
public class Solution {
    public static void main(String[] args) {
        short x = 10;
        x = x * 5;
        System.out.println(x);
    }
}
```

(a) 50 (b) 10 (c) 15 (d) 0 (e) Compile error

Study the diagram below and answer questions 32 and 33



32. How many variables does the Director class have? (a) 1 (b) 3 (c) 4 (d) 5 (e) 7

33. If the class defines a method, getDetails(), which of the following an overridden method?

(a) Engineer.getDetails();
(b) Director.getDetails();
(c) Employee.getDetails();
(d) Secretary.getDetails();
(e) None of the Above

OPTION B

34. The values of an object's attributes are known as its _____. (a) state (b) orientation (c) method (d) condition (e) class
35. An instance of a class is an _____. (a) method (b) procedure (c) object (d) case (e) class
36. Java is architecturally _____. (a) neutral (b) oriented (c) specifics (d) abstract (e) object
37. All Java programming statements must end with a _____. (a) period (b) comma (c) closing parenthesis (d) semicolon (e) period
38. Arguments to methods always appear within _____. (a) parentheses (b) double quotation marks (c) single quotation marks (d) curly braces (e) square bracket
39. In a Java program, you must use _____ to separate classes, objects, and methods. (a) commas (b) semicolons (c) dots (d) forward slashes (e) apostrophe
40. Languages that let you use an easily understood vocabulary of descriptive terms, such as read, write, or add, are known as _____ languages.
- (a) procedural (b) machine (c) object-oriented (d) high-level (e) structured
41. The rules of a programming language constitute its _____. (a) logic (b) syntax (c) format (d) objects (e) byte code
42. A _____ translates high-level language statements into machine code. (a) programmer (b) syntax detector (c) compiler (d) decipherer (e) code
43. Named computer memory locations are called _____. (a) compilers (b) addresses (c) appellations (d) variables (e) bus
44. Envisioning program components as objects that are similar to concrete objects in the real world is the hallmark of _____. (a) command-line operating systems (b) procedural programming (c) object-oriented programming (d) machine languages (e) method
45. All Java applications must have a method named _____. (a) method () (b) main () (c) java () (d) Hello () (e) class ()



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

COURSE: IFT211 - DIGITAL LOGIC AND DESIGN

Time: 60 Minutes

INSTRUCTIONS: Attempt all Questions and shade the correct option corresponding to the answer chosen.

1. Which gate is known as the universal gate? (a) NAND (b) OR (c) AND (d) XOR (e) None of the above
2. How does a logic gate function? (a) Using the flow of electric current (b) Spontaneously (c) By using energy from a chemical reaction (d) All of the above (e) None of the above
3. Which of the following is not a logic gate? (a) AND (b) OR (c) IF (d) NOT (e) All of the above
4. Which of the following signifies a NOT gate? (a) 1 (input)- 0 (output) (b) 1,0 (input)- 1 (output) (c) 0,0 (input)- 0 (output) (d) 1,1(input)-0 (output) (e) None of the above
5. Which of the following gates can function on a single input? (a) NOT (b) AND (c) OR (d) NOR (e) None of the above
6. A NAND gate has — inputs and — outputs. (a) High input and High output (b) High input and low output (c) Low input and low output (d) Low input and high output (e) None of the above
7. A Truth Table is a ... (a) table representing the output one will get for a given input (b) table representing what input and output will be given by a given boolean operator (c) mathematics table used to determine the output of logic (d) fundamental tool in digital electronics, computer science, and logic (e) All of the above.
8. How can a full adder be implemented using half adders? (a) 1 half adder (b) 2 half adders and 1 OR gate (c) 3 half adders and 2 OR gates (d) Only with full adders (e) None of the above
9. What combination is a NAND gate? (a) NOT AND (b) NOT OR (c) NOT NOT (d) XOR NOR (e) None of the above
10. A minterm is defined as (a) A product of all variables in complemented form (b) A product of all variables in either complemented or uncomplemented form (c) sum of all variables in complemented form (d) sum of all variables in either complemented or uncomplemented form (e) None of the above
11. A maxterm is defined as the (a) sum of all variables in either complemented or uncomplemented form (b) product of all variables in complemented form (c) product of all variables in either complemented or uncomplemented form (d) sum of all variables in uncomplemented form only (e) All of the above
12. The minterm for the combination $A = 1, B = 0, C = 1$ is: (a) $A + B' + C$ (b) $A'B'C'$ (c) $AB'C$ (d) ABC (e) None of the above
13. The maxterm for the combination $A = 1, B = 0$ is: (a) $A + B$ (b) $A + B'$ (c) $A' + B$ (d) $A' + B'$ (e) None of the above
14. Which of the following represents a canonical sum of products (SOP) form? (a) $AB + A'B'$ (b) $AB + AC$ (c) $A + B + C$ (d) $A'B'C + AB'C + ABC$ (e) None of the above
15. Which of the following is a sequential circuit? (a) Multiplexer (b) Decoder (c) Flip-Flop (d) Adder (e) None of the above
16. What distinguishes a sequential circuit from a combinational circuit? (a) Use of only logic gates (b) Dependence on current inputs only (c) Dependence on both current inputs and previous states (d) It has no outputs (e) None of the above
17. Which memory element is commonly used in sequential circuits? (a) Resistor (b) Capacitor (c) Flip-Flop (d) Inductor (e) None of the above
18. What type of sequential circuit does not use a clock signal? (a) Synchronous (b) Asynchronous (c) Combinational (d) Mixed (e) None of the above

19. The output of a flip-flop is dependent on: (a) Only the input at that time (b) The input and the previous output (c) The clock frequency only (d) Only the previous output (e) None of the above
20. What is the main drawback of an SR flip-flop? (a) It cannot store data (b) It has a forbidden input condition (c) It operates without a clock (d) It consumes too much power (e) None of the above
21. Which of the following flip-flops is free from the invalid state? (a) SR flip-flop (b) JK flip-flop (c) D flip-flop (d) T flip-flop (e) None of the above
22. In a D flip-flop, what happens when the clock triggers? (a) Q becomes 0 (b) Q becomes 1 (c) Q copies the D input (d) Q toggles (e) None of the above
23. What is the function of the clock signal in a sequential circuit? (a) To process inputs (b) To maintain voltage (c) To synchronize changes in the circuit (d) To generate outputs (e) None of the above
24. What is a combinational circuit? (a) A circuit with feedback loops (b) A circuit whose output depends only on the present inputs (c) A circuit that stores previous states (d) A sequential circuit (e) None of the above
25. A full adder has how many inputs and outputs? (a) 3 inputs, 2 outputs (b) 2 inputs, 1 output (c) 3 inputs, 1 output (d) 2 inputs, 2 outputs (e) None of the above
26. Which gate is used in the design of a half adder? (a) AND and OR gates (b) OR and XOR gates (c) XOR and AND gates (d) NAND and XOR gates (e) None of the above
27. Which of the following is NOT a combinational circuit? (a) Decoder (b) Encoder (c) Flip-Flop (d) Multiplexer (e) None of the above
28. Which of the following is the identity law in Boolean algebra? (a) $A + 0 = A$ (b) $A + A = A$ (c) $A + 1 = 1$ (d) $A \cdot A = A$ (e) None of the above
29. What is the value of $A + A'$? (a) A (b) 0 (c) 1 (d) A' (e) None of the above
30. Which Boolean expression represents the AND operation? (a) $A + B$ (b) $A \cdot B$ (c) $A \oplus B$ (d) A' (e) None of the above
31. What is $A \cdot 0$ equal to? (a) 0 (b) 1 (c) A (d) A' (e) None of the above
32. What is the dual of the Boolean expression $A + 0 = A$? (a) $A \cdot 0 = A$ (b) $A \cdot 1 = A$ (c) $A + 1 = A$ (d) $A + A = A$ (e) None of the above
33. What is the binary equivalent of the decimal number 13? (a) 1100 (b) 1011 (c) 1101 (d) 1110 (e) None of the above
34. What is the Boolean expression for a NAND gate? (a) $A + B$ (b) $A \cdot B$ (c) $(A \cdot B)'$ (d) $A \oplus B$ (e) None of the above
35. Which of the following is a basic logic gate? (a) AND (b) OR (c) NOT (d) All of the above (e) None of the above
36. What is the result of the binary addition $1011 + 1101$? (a) 11000 (b) 10100 (c) 10000 (d) 11100 (e) None of the above
37. Which gate gives a high output of (1) when any one of its inputs is high? (a) AND (b) OR (c) NOT (d) NAND (e) None of the above
38. Which gate is a NOT universal gate? (a) RAND (b) ORAND (c) NORAND (d) NOTAND (e) All of the above
39. How many inputs and outputs does a half adder have? (a) 1 input, 2 outputs (b) 2 inputs, 1 output (c) 2 inputs, 2 outputs (d) 3 inputs, 2 outputs (e) None of the above
40. Which of the following limitations does a half adder have? (a) Cannot add binary numbers (b) Cannot add two bits (c) Cannot handle carry from a previous stage (d) All of the above (e) None of the above

40

OR AND NOT

1000 1001 1010 1011 1100 1101



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

Course Code: DTS201 - Introduction to Data Science

CREDIT UNIT: 02

TIME: One Hour

Instruction: Answer ALL Questions

Shade right options with HB Pencil

Section A: Fundamentals of Data Science

1. Which of the following best describes the primary goal of data science?
a) Data storage optimization
b) Developing user interfaces
c) ☒ Extracting insights from data
d) Designing hardware components
e) Managing network protocols
2. In data science, the term "feature" refers to:
a) A software bug
b) ☒ An individual measurable property of a phenomenon
c) A type of data visualization
d) A data storage unit
e) A machine learning algorithm
3. Which discipline is NOT traditionally integrated into data science?
a) Statistics
b) Computer Science
c) Domain Expertise
d) ☒ Astrology
e) Mathematics
4. The process of cleaning and organizing raw data into a usable format is known as:
a) Data visualization
b) Data mining
c) ☒ Data preprocessing
d) Data warehousing
e) Data encryption
5. Which of the following is a characteristic of big data?
a) Low volume
b) Static structure
c) ☒ High velocity
d) Limited variety
e) Minimal veracity
6. The term "data science" was first coined in which decade?
a) 1960s
b) ☒ 1970s
c) 1980s
d) 1990s
e) 2000s
7. Which of the following is NOT a step in the data science process?
a) Data collection
b) Data preprocessing
c) Model deployment
d) ☒ Software engineering
e) Model evaluation
8. In the context of data science, "EDA" stands for:
a) Electronic Data Analysis
b) ☒ Exploratory Data Analysis
c) Enhanced Data Algorithm
d) External Data Access
e) None of the above
9. Which of the following best describes supervised learning?
a) Learning without labeled data
b) ☒ Learning with labeled data
c) Learning through reinforcement

- d) Learning via unsupervised methods
- e) None of the above
10. Which of the following is an example of unstructured data?
a) Excel spreadsheet
b) SQL database
c) ☒ Text document
d) CSV file
e) Relational table

Section B: Data Extraction and Big Data

11. Which of the following tools is commonly used for web scraping?
a) NumPy
b) ☒ BeautifulSoup
c) Matplotlib
d) TensorFlow
e) PyTorch
12. In big data, the "3 Vs" stand for Volume, Velocity, and:
a) Variety
b) Veracity
c) Validity
d) Value
e) ☒ Visualization
13. Which of the following is a distributed computing framework for big data processing?
a) Hadoop
b) Excel
c) SPSS
d) Tableau
e) SAS
14. MapReduce is a programming model associated with:
a) Data visualization
b) Big data processing
c) ☒ Statistical analysis
d) Machine learning
e) Database management
15. Which of the following is NOT a characteristic of big data?
a) Volume
b) Velocity
c) Variety
d) Veracity
e) ☒ Vagueness
16. The process of identifying patterns in large datasets to predict future trends is known as:
a) Data cleaning
b) ☒ Data mining
c) Data entry
d) Data storage
e) Data encryption
17. Which of the following is a NoSQL database?
a) MySQL
b) PostgreSQL
c) ☒ MongoDB
d) Oracle
e) SQLite
18. In the context of big data, "HDFS" stands for:
a) High-Definition File System
b) ☒ Hadoop Distributed File System
c) Hybrid Data File Storage
d) Hierarchical Data File System
e) None of the above
19. Which of the following is a real-time data processing framework?
a) Hadoop MapReduce
b) ☒ Apache Spark
c) Microsoft Excel
d) Tableau
e) SPSS

40. Which of the following is NOT a data extraction technique?
- a) Web scraping
 - b) API integration
 - c) Data entry
 - d) Data visualization
 - e) Database querying

Section C: Data Science Tools and Platforms

21. Which programming language is widely used for statistical analysis and data visualization?
- a) Java
 - b) C++
 - c) R
 - d) Swift
 - e) PHP
22. Which Python library is primarily used for data manipulation and analysis?
- a) NumPy
 - b) Pandas
 - c) Matplotlib
 - d) Seaborn
 - e) Scikit-learn
23. Which of the following tools is used for creating interactive data visualizations?
- a) Jupyter Notebook
 - b) Tableau
 - c) Notepad++
 - d) Eclipse
 - e) Git
24. Scikit-learn is a Python library used for:
- a) Web development
 - b) Machine learning
 - c) Data visualization
 - d) Database management
 - e) Network security
25. Which of the following is a cloud-based platform for data science and machine learning?
- a) Google Colab
 - b) Microsoft Word
 - c) Adobe Photoshop
 - d) VLC Media Player
 - e) Skype
26. Which of the following is NOT a Python library?
- a) NumPy
 - b) Pandas
 - c) ggplot2
 - d) Matplotlib
 - e) Seaborn
27. Which tool is commonly used for version control in data science projects?
- a) Git
 - b) Excel
 - c) Tableau
 - d) PowerPoint
 - e) SPSS
28. Which of the following platforms provides a collaborative environment for data science projects?
- a) GitHub
 - b) Dropbox
 - c) OneDrive
 - d) Google Drive
 - e) iCloud
29. Which of the following is a Python-based web framework?
- a) Django
 - b) Flask
 - c) Both a and b
 - d) React
 - e) Angular
30. Which of the following is used for creating dashboards and business intelligence reports?
- a) Tableau
 - b) Excel
 - c) Power BI
 - d) All of the above
 - e) None of the above

Section D: Data Science Processes and Methodologies

31. The process of dividing data into training and testing sets is known as:
- a) Data splitting
 - b) Data shuffling

- c) Data merging
 - d) Data cleaning
 - e) Data visualization
32. Which of the following is NOT a step in the CRISP-DM process model?
- a) Business understanding
 - b) Data understanding
 - c) Data preparation
 - d) Model deployment
 - e) Data encryption
33. In data preprocessing, handling missing values can be done by:
- a) Deleting rows with missing values
 - b) Replacing missing values with mean or median
 - c) Using algorithms that support missing values
 - d) All of the above
 - e) None of the above
34. Which of the following is a technique for reducing the dimensionality of data?
- a) Principal Component Analysis (PCA)
 - b) Linear Regression
 - c) Decision Trees
 - d) K-Means Clustering
 - e) Naive Bayes
35. Which of the following is a measure of model performance for classification problems?
- a) Mean Squared Error
 - b) R-squared
 - c) Accuracy
 - d) Adjusted R-squared
 - e) None of the above
36. Which of the following is NOT a data preprocessing technique?
- a) Normalization
 - b) Standardization
 - c) Data augmentation
 - d) Data encryption
 - e) Handling missing values
37. The process of transforming raw data into an understandable format is known as:
- a) Data cleaning
 - b) Data preprocessing
 - c) Data transformation
 - d) Data integration
 - e) Data visualization
38. Which of the following is a technique used to prevent overfitting in machine learning models?
- a) Cross-validation
 - b) Underfitting
 - c) Data leakage
 - d) Data augmentation
 - e) None of the above
39. Which of the following is NOT a type of machine learning?
- a) Supervised learning
 - b) Unsupervised learning
 - c) Reinforcement learning
 - d) Deep learning
 - e) None of the above
40. Which of the following is a key difference between data analytics and data science?
- a) Data analytics is more focused on future predictions, while data science focuses on historical analysis.
 - b) Data science is a broader field that includes data analytics, along with aspects of machine learning, statistics, and computer science.
 - c) Data analytics deals with structured data, while data science handles unstructured data.
 - d) Data science is primarily used in academic research, while data analytics is used in business.
 - e) There is no significant difference between data analytics and data science.

B



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W/26 vs 11 E, 12

Option A

SECOND SEMESTER B.Sc. (FULL-TIME) EXAMINATIONS 2024/2025 SESSION

COURSE TITLE: Elementary Differential Equations (MTH202)

TIME ALLOWED: 1 Hr 45 Mins

INSTRUCTIONS: (i) Write and circle your attendance list serial number on the objective answer paper.
(ii) Attempt all questions by SHADING (using HB pencil) the letter box that corresponds to the correct option.
Information about your Mat. No., Name, Course code, Faculty code and Departmental code must be clearly written and CORRECTLY SHADED. YOU MUST SUBMIT YOUR QUESTION PAPER ALONG WITH YOUR ANSWER SHEET.

1. The order and degree of the differential equation $\left(\frac{dy}{dx}\right)^3 + y^2 = 8$ are respectively;
(a) 1, 1 (b) 2, 3 (c) 1, 3 (d) 3, 3 (e) 1, 2
2. The order and degree of the differential equation $\sec x = y\sqrt{\frac{dy}{dx}}$ are respectively;
(a) 2, 1 (b) 1, 2 (c) 2, 2 (d) 1, 1 (e) 1, 3
3. The order and degree of the differential equation $\left(1 + \frac{dy}{dx}\right)^{\frac{2}{3}} = e^{\frac{d^2y}{dx^2}}$ are respectively
(a) 2, 1 (b) 3, 5 (c) 2, 3 (d) 1, 3 (e) 2, 5
4. The order and degree of the differential equation $\ddot{x} + \log(1+t)\dot{x} = 0$ are respectively
(a) 1, 1 (b) 2, 1 (c) 2, 2 (d) 3, 1 (e) 1, 2
5. Determine the differential equation whose general solution is $y = cx^4$
(a) $\frac{dy}{dx} = 4x^3$ (b) $x\frac{dy}{dx} = y$ (c) $\frac{dy}{dx} = \frac{4y}{x}$ (d) $\frac{dy}{dx} = \frac{x}{y}$ (e) $x^2\frac{dy}{dx} = 4y$
6. Determine the differential equation whose general solution is $y = ce^{3x}$
(a) $\frac{dy}{dx} + 3y = 0$ (b) $\frac{dy}{dx} - y = 0$ (c) $\frac{dy}{dx} = 3y$ (d) $\frac{dy}{dx} = \frac{y}{3}$ (e) $\frac{dy}{dx} = 4y$
7. Determine the differential equation whose general solution is $y = A\cos x + B\sin x$
(a) $y'' + y = 0$ (b) $y' + y = 0$ (c) $y'' - y = 0$ (d) $y' - y = 0$ (e) $y'' + 2y = 0$
8. Solve the differential equation $4ydx + xdy = 0$
(a) $yx^3 = c$ (b) $x^4y = c$ (c) $yx = c$ (d) $xy^4 = c$ (e) $xy^2 = c$
9. Solve the differential equation $t\frac{dy}{dt} + 5y = 0, y(1) = 1$
(a) $y = t^5$ (b) $y = \frac{5}{t}$ (c) $y = \frac{1}{t^5}$ (d) $y = 5t$ (e) $y = t^2$
10. Find the particular solution of the equation $e^x \frac{dy}{dx} = 4$ given that $y = 3$, when $x = 0$
(a) $y = -4e^{-x} + 7$ (b) $y = 4e^{-x} + 7$ (c) $y = 4e^x + 7$ (d) $y = 4e^x + 7$ (e) $y = e^x + 7$
11. Solve the differential equation $x\frac{dy}{dx} - 3y = 4x$
(a) $y = Cx^3 - 2x$ (b) $y = Ce^{3x} - 2x$ (c) $y = x + C$ (d) $y = 2x + Cx^3$ (e) $y = -\frac{2}{x} + Cx^3$
12. Which of the following is a First Order Homogeneous Ordinary Differential Equation?
(a) $\frac{dy}{dx} = x^2 + y$ (b) $\frac{dy}{dx} = y + x^2$ (c) $\frac{dy}{dx} = xy$ (d) $\frac{dy}{dx} = x + y^2$ (e) $\frac{dy}{dx} = \frac{x+y}{x}$

13. Which of the following is NOT a First Order Homogeneous Ordinary Differential Equation?

(a) $\frac{dy}{dx} = \frac{y}{x}$ (b) $\frac{dy}{dx} = \frac{x+y}{x}$ (c) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$ (d) $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$ (e) $\frac{dy}{dx} = x + y$

14. Which of the following is a First Order Exact Ordinary Differential Equation? (a) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$ (b) $(2x^3y)dx + (y^3 + x)dy = 0$ (c) $(ye^x + x)dx + (xe^x + y)dy = 0$ (d) $(2xy^2 + y)dx + (x^2y + x)dy = 0$ (e) $(3x + 2y)dx + (6x + 4y)dy = 0$

15. Which of the following is NOT a First Order Exact Ordinary Differential Equation? (a) $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$ (b) $(xy + 1)dx + (\frac{1}{2}x^2 + y)dy = 0$ (c) $(ye^x + x)dx + (xe^x + y)dy = 0$ (d) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$ (e) $(3x^2y + 2y)dx + (x^3 + 2x)dy = 0$

16. For the following general solution, obtain a particular solution using the given conditions. $y = A\cos 2x + B\sin 2x$, $y(\frac{\pi}{4}) = 3$, and $y'(\frac{\pi}{4}) = 8$. (a) $y = 4\cos 2x + 3\sin 2x$ (b) $y = -4\cos 2x + 3\sin 2x$ (c) $y = -4\cos 2x - 3\sin 2x$ (d) $y = 4\cos 2x - 3\sin 2x$ (e) None of the above

17. Determine the differential equation whose general solution is $y = Ae^{2x} + Be^{-x}$ (a) $y'' + y' - 2y = 0$ (b) $y'' - y' + 2y = 0$ (c) $y'' - y' - 2y = 0$ (d) $y'' + y' + 2y = 0$ (e) None of the above

18. Solve $\frac{dy}{dx} = \frac{x+y}{2x}$ (a) $(x+y)^2 = Ax^3$ (b) $(1+y)^2 = Ax^3$ (c) $(x+y)^2 = Ax^2$ (d) $(1+y)^2 = Ax^3$ (e) None of the above

19. Solve $x\frac{dy}{dx} - 5y = x^7$ (a) $y = \frac{x^2}{2} + cx^7$ (b) $y = \frac{x^7}{2} + cx^5$ (c) $y = \frac{x^5}{2} - cx^7$ (d) $y = \frac{x^7}{2} - cx^5$ (e) None of the above

20. Solve $x\frac{dy}{dx} - 3y = 4x$ (a) $y = -x + cx^2$ (b) $y = -2x + cx^3$ (c) $y = -\frac{2}{5}x + cx^5$ (d) $y = -5x + cx^3$ (e) None of the above

21. Solve $x\frac{dy}{dx} - 4y = x^3$ (a) $y = Cx^2 - x^3$ (b) $y = Cx^4 + x^3$ (c) $y = Cx^4 - x^3$ (d) $y = Cx^3 - x^4$ (e) None of the above

22. Solve the homogenous differential equation $(x+2y)dx + (2x+3y)dy = 0$ (a) $x^2 + 4xy + 3y^2 = c$ (b) $x^2 - 2xy - 3y^2 = c$ (c) $x^2 - 4xy + y^2 = c$ (d) $x^2 - xy - y^2 = c$ (e) None of the above

23. Solve $y' + y = e^x$ (a) $y = 2e^{-x} + Ce^x$ (b) $y = \frac{1}{2}e^{-x} + Ce^x$ (c) $y = e^x + Ce^{-x}$ (d) $y = \frac{1}{2}e^x + Ce^{-x}$ (e) None of the above

24. Solve $\frac{dy}{dx} + \frac{1}{x}y = xy^2$ (a) $y = (cx - x^2)^{-2}$ (b) $y = (x^2 - cx)^{-2}$ (c) $y = (x^2 - cx)^{-1}$ (d) $y = (cx - x^2)^{-1}$ (e) None of the above

25. Solve $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = 0$, $y(0) = 9$, $y'(0) = 5$ (a) $e^{-3t} + 8e^t$ (b) $\beta_1 e^{-3t} + \beta_2 e^t$ (c) $\beta_1 e^{-3t} + \beta_2 e^{-t}$ (d) $e^{-3t} + 8e^{-t}$ (e) None of the above

26. Given that $A\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Cy = 0$, If the discriminant, $B^2 - 4AC > 0$. What is the possible general solution if the roots are m_1 and m_2 ? (a) $\beta_1 e^{m_1 x} + \beta_2 e^{m_2 x}$ (b) $\beta_1 e^{m_1 x} + \beta_2 x e^{m_1 x}$ (c) $e^{m_1 x}(\beta_1 \sin m_2 x + \beta_2 \cos m_2 x)$ (d) $\beta_1 + \beta_2(e^{m_1 x} + e^{m_2 x})$ (e) None of the above

27. The second order differential equation $A \frac{d^2y}{dt^2} + B \frac{dy}{dt} + Cy = f(t)$ is said to be homogeneous if (a) $f(t) \neq 0$ (b) $f(t) > 0$ (c) $f(t) = 0$ (d) $f(t) = 0$ or $f(t) > 0$ (e) None of the above ✓
28. Solve $\frac{d^2y}{dt^2} - 2 \frac{dy}{dt} + 5y = 0$, $y(0) = 5$, $y'(0) = 5$ (a) $e^t(5\cos 2t + 2\sin 2t)$ (b) $e^{2t}(5\cos 2t + 2\sin 2t)$ (c) $e^t(5\cos 2t - 2\sin 2t)$ (d) $e^{2t}(5\cos 2t - 2\sin 2t)$ (e) None of the above ✓
29. Solve $\frac{d^2y}{dt^2} + 9 \frac{dy}{dt} + 20y = f(t)$ where $f(t) = 11e^{-4t}$ (a) $11te^{-4t}$ (b) $11te^{-4t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$ (c) $10e^{3t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$ (d) $2e^{3t} + \beta_1 e^{-5t} + \beta_2 e^{4t}$ (e) None of the above ✗
30. The solution to the inhomogeneous part (or the particular solution, y_p) of the differential equation, $\frac{d^2y}{dt^2} + 2 \frac{dy}{dt} - 3y = f(t)$ where $f(t) = \sin 3t$ is (a) $\frac{1}{30}(\sin 3t + 2\cos 3t)$ (b) $-\frac{1}{30}(2\sin 3t - \cos 3t)$ (c) $-\frac{1}{30}(2\sin 3t + \cos 3t)$ (d) $\frac{1}{30}(\sin 3t + 4\cos 3t)$ (e) None of the above ✓
31. A particle moves in a straight line such that for a short time its speed is defined by $v = (3t^2 + 4t)$ m/s where t is in seconds. Determine the distance (s) of the particle when $t = 5$ seconds with initial value of $t = 0$ and $s = 0$ (a) 175m (b) 200m (c) 400m (d) 800m (e) None of the above ✓
32. A particle moves on a straight line from a position $s = 5t^3 + 3t^2$ in metre at time t . Determine the acceleration of the particle at time $t = 3$ sec. (a) 14 ms^{-2} (b) 96 ms^{-2} (c) 100 ms^{-2} (d) 90 ms^{-2} (e) None of the above ✗
33. A car moves in a three-dimensional space such that for a short time its speed is given by $v = (4t + 5, 2t + 3t^3, t + 5t^2) \text{ ms}^{-1}$ where t is the time in seconds. Determine its position in 2 seconds. (a) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix} \text{ m}$ (b) $\begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \text{ m}$ (c) $\begin{pmatrix} 12 \\ 15 \\ 2 \end{pmatrix} \text{ m}$ (d) $\begin{pmatrix} 18 \\ 16 \\ 46 \\ 3 \end{pmatrix} \text{ m}$ (e) None of the above ✗
34. Solve $\frac{d^2y}{dt^2} + 9 \frac{dy}{dt} + 20y = 0$ (a) $\beta_1 e^{-5t} + \beta_2 e^{-4t}$ (b) $\beta_1 e^{-4t} + \beta_2 e^{5t}$ (c) $\beta_1 e^{5t} + \beta_2 e^{6t}$ (d) $\beta_1 e^{4t} + \beta_2 e^{5t}$ (e) None of the above ✓
35. Solve $y'' + 4y' + 3y = 15$ (a) $\beta_1 e^{-2x} + \beta_2 e^{-4x} + 5x$ (b) $\beta_1 e^{2x} + \beta_2 e^{-4x} + 15$ (c) $\beta_1 e^{-2x} + \beta_2 e^{-x} + 5$ (d) $\beta_1 e^{3x} + \beta_2 e^{-x} + 15x$ (e) None of the above ✗



DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF PHYSICAL SCIENCES
UNIVERSITY OF BENIN, BENIN CITY

2024/2025 BSC COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS

COURSE: CYB201 - CYBERSECURITY AND STRATEGY

CREDIT UNIT: 02

TIME: 50 MINUTES.

INSTRUCTION: Answer all the Questions by shading the corresponding correct answers on YOUR OMR sheet.

1. What does the CIA Triad stand for in information security? A) Confidentiality, Integrity, Accessibility B) Control, Integrity, Availability C) Confidentiality, Integrity, Availability D) Compliance, Integrity, Authentication E) Confidentiality, Identification, Availability
2. Which encryption standard uses the same key for encryption and decryption? A) RSA B) AES C) SHA-256 D) TLS E) OAuth
3. A hospital uses AES encryption for patient records and maintains offline backups. Which principle of the CIA Triad does this primarily address during a ransomware attack? A) Confidentiality B) Integrity C) Availability D) Authentication E) Non-repudiation
4. Data classified as "Secret/Restricted" requires which security measure? A) Basic access controls B) Public sharing on the company website C) Multi-factor authentication and isolated networks D) TLS for data in transit only E) Weekly vulnerability scans
5. Which method ensures data on a decommissioned hard drive is irretrievable? A) Formatting the drive B) Cryptographic erasure C) Using a firewall D) Installing antivirus software E) Enabling MFA
6. Which protocol is used for enterprise Single Sign-On (SSO)? A) HTTP B) SAML C) WPA3 D) AES E) DNSSEC
7. A penetration tester with no prior knowledge of the target system is performing: A) White Box Testing B) Grey Box Testing C) Black Box Testing D) SAST E) DAST
8. Phishing simulation exercises aim to: A) Test network bandwidth B) Train employees to recognize social engineering attacks C) Encrypt sensitive emails D) Update antivirus signatures E) Generate threat intelligence reports
9. A VLAN is used for: A) Network segmentation B) Encrypting emails C) Multi-factor authentication D) Generating hashes E) Backing up data
10. A vendor security assessment is part of: A) Third-Party Risk Management (TPRM) B) Incident Response C) Cryptographic erasure D) Data classification E) TLS implementation
11. Security awareness training reduces risks associated with: A) Hardware failures B) Social engineering C) AES key management D) Firewall misconfigurations E) Data redundancy
12. A Web Application Firewall (WAF) protects against: A) SQL injection B) Data theft from decommissioned drives C) Insider threats D) Weak encryption keys E) Phishing emails
13. STRIDE is a methodology for: A) Threat modeling B) Calculating ALE C) Securing APIs D) Network segmentation E) Data classification
14. JWT (JSON Web Tokens) are often used for: A) API authentication B) Encrypting databases C) Securing DNS D) Data erasure E) Security awareness
15. Which control ensures users have minimal necessary access? A) Least Privilege B) Defense in Depth C) Redundancy D) Hashing E) Salting
16. A file's SHA-256 hash changes if: A) Its name is modified B) Its content is altered C) It is moved to a new folder D) It is encrypted with AES E) It is uploaded to the cloud
17. Which protocol is used for secure remote server access? A) SSH B) HTTP C) DNS D) FTP E) SNMP
18. An IPS (Intrusion Prevention System) can: A) Block malicious traffic in real-time B) Encrypt data at rest C) Generate threat intelligence reports D) Classify data E) Train employees
19. "Data at rest" refers to information: A) Transmitted over a network B) Stored on a hard drive C) Processed in memory D) Displayed on a screen E) Deleted via cryptographic erasure
20. A phishing simulation targets: A) Network bandwidth B) Employee vigilance C) Firewall rules D) Data classification E) Backup integrity
21. Which of the following is NOT one of the three categories of cybersecurity methods mentioned in the notes? A) Preventive measures B) Detective measures C) Corrective measures D) Predictive measures E) All of the above are cybersecurity methods
22. What is the main difference between a virus and a worm? A) Viruses are more dangerous than worms B) Worms are larger in file size than viruses. C) Viruses require a host program to execute, worms do not D) Worms only affect Windows systems, viruses affect all systems E) There is no difference between viruses and worms
23. Which fault-tolerant methodology involves using different types of components for protection? A) Redundancy

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B) Diversity C) Isolation D) Authentication E) Non-repudiation

24. What does "non-repudiation" mean in cybersecurity? A) Preventing unauthorized access to information B) Ensuring authorized users have access when needed C) Verifying the identity of users or devices D) Ensuring users cannot deny having performed an action E) Protecting information from unauthorized modification
25. Which of the following is an example of the "availability" cybersecurity element? A) A company encrypts its sensitive data B) A bank uses digital signatures for transactions C) A website uses a CDN to handle large numbers of visitors D) A user enters username and password to log in E) A company uses a firewall to block unauthorized access
26. The WannaCry ransomware attack infected approximately how many computers and in how many countries? A) Over 100,000 computers in 100 countries B) Over 200,000 computers in 150 countries C) Over 300,000 computers in 200 countries D) Over 50,000 computers in 75 countries E) Over 500,000 computers in 250 countries
27. What is the difference between authentication and authorization? A) Authentication verifies identity, authorization grants or denies access to resources B) Authorization verifies identity, authentication grants or denies access to resources C) They are the same thing with different names D) Authentication is for users, authorization is for devices E) Authentication is preventive, authorization is detective
28. Which of the following is NOT a direct advantage of cybersecurity in an organization? A) Protects sensitive information B) Maintains the integrity of data C) Ensures the availability of systems and data D) Builds trust with customers and partners E) Reduces operational costs for organizations
29. Which type of security policy defines appropriate use of company resources, including computers, internet, and email? A) Access Control Policy B) Acceptable Use Policy (AUP) C) Information Security Policy D) Incident Response Policy E) Data Protection & Encryption Policy
30. What are the six phases of the NIST Incident Response Lifecycle in the correct order? A) Detection, Preparation, Containment, Eradication, Recovery, Lessons Learned B) Preparation, Detection & Analysis, Containment, Eradication, Recovery, Lessons Learned C) Containment, Detection, Preparation, Eradication, Recovery, Analysis D) Preparation, Containment, Detection, Recovery, Eradication, Lessons Learned E) Detection, Containment, Preparation, Recovery, Eradication, Lessons Learned
31. In the Target Data Breach (2013), what was the primary security failure that led to the compromise? A) Employees clicked on phishing emails B) The company failed to encrypt customer data C) Target failed to enforce strict third-party access controls D) The company used weak passwords E) Target's firewall was misconfigured
32. Which cybersecurity best practice principle states that employees should have the minimum access necessary for their jobs? A) Multi-Factor Authentication (MFA) B) Defense in Depth C) Principle of Least Privilege (PoLP) D) Zero Trust Architecture E) Security by Design
33. What type of security testing involves ethical hackers simulating real-world cyberattacks to test system defenses?
A) Vulnerability Scanning B) Security Audits C) Code Review D) Penetration Testing (Ethical Hacking) E) Social Engineering Testing
34. Which security testing method uses automated tools to detect system vulnerabilities? A) Penetration Testing B) Vulnerability Scanning C) Security Audits D) Social Engineering Testing E) Code Review
35. software vulnerabilities C) Improper enforcement of multi-factor authentication (MFA) D) Lack of employee training E) Misconfigured firewall rules to Weak Passwords Default
36. Which of the following may NOT be listed as a key cybersecurity best practice? A) Multi-Factor Authentication (MFA) B) Regular Security Updates & Patch Management C) Employee Cybersecurity Training D) Blockchain Implementation E) Principle of Least Privilege (PoLP)
37. According to the NIST Incident Response Lifecycle, what is the correct order of phases? A) Detection, Preparation, Containment, Recovery, Eradication, Lessons Learned B) Preparation, Detection & Analysis, Containment, Eradication, Recovery, Lessons Learned C) Containment, Detection, Preparation, Eradication, Recovery, Analysis D) Preparation, Containment, Detection & Analysis, Recovery, Eradication, Lessons Learned E) Detection & Analysis, Preparation, Containment, Recovery, Eradication, Lessons Learned
38. In the Colonial Pipeline Ransomware Attack (2021), what was the primary security weakness that allowed the attackers to gain access? A) Unpatched software vulnerabilities B) A compromised single employee password C) Weak firewall configuration D) Lack of network segmentation E) Inadequate backup systems
39. What are the three main goals of cybersecurity? A) Prevention, Detection, Response B) Confidentiality, Integrity, Availability C) Authentication, Authorization, Accounting D) Encryption, Firewalls, Antivirus E) none of the above
40. The three main types of access control are A) Mandatory access control, Discretionary main control and Role-play access control B) Discretionary main control, Mandatory based control, and Role-based access control C) Mandatory access control, Integrity based Control and Discretionary based control D) Mandatory Access based control, Role-based Access Control and Discretionary access control E) all of the above

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2024/2025 BSc COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS

COURSE: SEN201 - INTRODUCTION TO SOFTWARE ENGINEERING

CREDIT UNIT: 03

OPTION C

TIME: 50 minutes

INSTRUCTION: Attempt all questions and shade on your OMR sheet, the option corresponding to the answer choice for each question.

1. What does the multiplicity 1..*, in an association imply? ☒ (a) Exactly one instance. (b) Zero or one instance. (c) One or more instances. (d) A fixed number of instances, like 5. (e) Zero or more instances.
2. Aggregation is a special form of association that represents _____. ☒ (a) A strong ownership and lifecycle dependency. (b) A whole-part relationship where the part can exist independently of the whole. (c) An inheritance hierarchy. (d) The sequence of interactions between objects. (e) The states an object can be in.
3. How is composition different from aggregation? ☒ (a) Composition is weaker than aggregation. (b) In composition, the part's lifecycle is independent of the whole. (c) Composition implies a stronger ownership where the part's lifecycle is dependent on the whole. (d) Composition is used only for database modeling. (e) There is no difference; the terms are synonymous.
4. Which relationship is represented by a diamond symbol in a Class Diagram? ☒ (a) Inheritance (Generalization) (b) Association (c) Dependency (d) Aggregation or Composition (e) Realization
5. The main advantage of using inheritance is: ☒ (a) It increases the number of objects. (b) It allows for easy modification and implementation reuse. (c) It maintains strict encapsulation. (d) It simplifies object creation at runtime. (e) It models dynamic relationships.
6. A Sequence Diagram is primarily used to show _____. ☒ (a) The static classes in the system. (b) The flow of activities across different swim lanes. (c) The interaction between objects in chronological order. (d) The states of an object over its lifetime. (e) The physical deployment of software components.
7. In a Sequence Diagram, what does a vertical dashed line represent? ☒ (a) A message between objects. (b) The lifetime of an object. (c) The activation period of an object. (d) A conditional branch. (e) The system boundary.
8. What is the key difference between a Sequence Diagram and a Collaboration Diagram? ☒ (a) Sequence Diagrams show structure, while Collaboration Diagrams show behavior. (b) Collaboration Diagrams show the sequence of messages using numbers, while Sequence Diagrams use vertical ordering. (c) Sequence Diagrams are obsolete and have been replaced by Collaboration Diagrams. (d) Collaboration Diagrams cannot show object creation. (e) There is no difference; they are identical.
9. An Activity Diagram is similar to a flowchart but is different because it supports _____. ☒ (a) The definition of classes and attributes. (b) The modeling of parallel activities and synchronization. (c) The showing of message sequences between objects. (d) The depiction of inheritance hierarchies. (e) The specification of operation signatures.
10. In an Activity Diagram, what do "swim lanes" help to illustrate?
☒ (a) The sequence of activities. (b) The conditions for transitions. (c) The grouping of activities based on who performs them. (d) The final state of the process. (e) The objects involved in the activities.

11. A State Chart Diagram is most useful for modeling one of the following: (a) The interactions between multiple objects in a use case. (b) The flow of activities in a business process. (c) How the state of a single object changes in response to events over its lifetime. (d) The static class structure. (e) The physical nodes where software is deployed.
12. In a State Chart Diagram, what is a "transition"? (a) A state that the object can be in. (b) The initial starting point of the object. (c) A change from one state to another triggered by an event. (d) The final state of the object. (e) An activity performed within a state.
13. The notation event [guard]/action on a transition in a State Chart Diagram means _____. (a) The action is performed before the event is checked. (b) The transition occurs if the event happens, regardless of the guard condition. (c) The transition occurs if the event happens and the guard condition is true, and then the action is performed. (d) The guard condition is evaluated only after the action is performed. (e) The action defines a new state for the object.
14. According to the document, the Classical Waterfall Model is considered "idealistic". What is its major shortcoming? (a) It is too expensive to implement. (b) It requires highly skilled project managers. (c) It assumes that each phase can be completed perfectly with no errors, which is impractical. (d) It does not allow for any customer interaction. (e) It is only suitable for small projects.
15. What is the main purpose of developing a prototype in software development? (a) To serve as the final, deployable version of the software. (b) To identify and clarify user requirements through iterative feedback. (c) To replace the need for an SRS document. (d) To perform extensive load testing. (e) To train end-users before the actual system is built.
16. The Spiral Model is called a "meta-model" because it _____. (a) Is the simplest model to understand and use. (b) Is only used for prototyping user interfaces. (c) Encompasses all other life cycle models and includes risk handling. (d) Focuses exclusively on coding and testing. (e) Is used only for object-oriented development projects.
17. The Evolutionary Model is also known as the _____. (a) Waterfall Model (b) Spiral Model (c) Successive Versions or Incremental Model (d) Prototype Model (e) Agile Model
18. Which life cycle model is most suitable for projects where requirements are not well understood? (a) Classical Waterfall Model (b) Iterative Waterfall Model (c) Prototyping Model (d) Evolutionary Model (e) Spiral Model
19. In the example of documenting the "withdraw cash" function from an ATM, the amount to be withdrawn must be: (a) Any positive number. (b) An integer value greater than 100 and less than 10,000 in multiples of 100. (c) A value less than 100. (d) A decimal value for precision. (e) Entered as a string.
20. The concept of "Conceptual Integrity" in an SRS document means that the document should be _____. (a) As long as possible. (b) Easy to understand and consistent. (c) Focused on implementation details. (d) Written for developers only. (e) Verified only at the end of the project.
21. In the context of UML, what is a model? (a) A fully functional version of the software. (b) A simplified representation of a system, focusing on important aspects. (c) The final user documentation. (d) The project plan and schedule. (e) The source code repository.
22. What does UML stand for? (a) Universal Modeling Logic (b) Unified Module Language (c) Unified Modeling Language (d) Universal Module Linker (e) Unified Method Library
23. Which of the following is a key reason for creating models in software engineering? (a) To replace the need for coding. (b) To eliminate the need for testing. (c) To manage complexity and facilitate communication. (d) To automatically generate user manuals. (e) To calculate the project's return on investment.

24. Which UML view represents the system's functionality as seen by external users? (a) Structural View (b) Behavioral View (c) Implementation View (d) Environmental View (e) User's View
25. The Use Case Diagram is part of which UML view? (a) Structural View (b) Behavioral View (c) User's View (d) Implementation View (e) Environmental View
26. In a Use Case Diagram, an actor represents: (a) A specific person who will use the system. (b) A role played by a user with respect to the system. (c) An internal class within the system. (d) A database table. (e) A specific scenario of system use
27. What is the purpose of the <<include>> relationship between use cases? (a) To show optional behavior under certain conditions. (b) To factor out common behavior that is used multiple times. (c) To indicate that one use case is a special case of another. (d) To represent the system boundary. (e) To model the inheritance relationship between actors.
28. What does the <<extend>> relationship in a Use Case Diagram signify? (a) Compulsory and automatic inclusion of one use case in another. (b) That an actor can participate in multiple use cases. (c) Optional behavior that is triggered under specific conditions. (d) The hierarchical decomposition of a complex use case. (e) The sequence of steps within a use case.
29. Which UML diagram describes the static structure of a system in terms of classes and their relationships? (a) Use Case Diagram (b) Sequence Diagram (c) Class Diagram (d) Activity Diagram (e) State Chart Diagram
30. In a Class Diagram, what does an association between two classes represent? (a) An inheritance relationship. (b) A communication path or connection between instances of the classes. (c) That one class is a part of another class. (d) The flow of activities from one class to another. (e) The sequence of messages exchanged.
31. What is the primary purpose of the Requirements Analysis and Specification phase? (a) Writing the source code for the software. (b) Understanding, defining, and documenting what a software system should do. (c) Testing the final software product for bugs. (d) Designing the user interface of the system. (e) Deploying the software to the end-users.
32. _____ are considered stakeholders in a software project? (a) Only the programmers and testers. (b) The users and clients. (c) Individuals or groups with an interest in or affected by the software system. (d) The project managers and business analysts. (e) Only the regulatory bodies.
33. What is the main difference between functional and non-functional requirements? (a) Functional requirements are about cost, while non-functional are about time. (b) Functional requirements are optional, while non-functional are mandatory. (c) Functional requirements specify what the system should do, while non-functional specify how well it should do it. (d) Functional requirements are technical, while non-functional are business-oriented. (e) There is no significant difference.
34. What document serves as a formal record of the analyzed requirements? (a) Project Charter (b) Test Case Document (c) Requirements Specification Document (RSD) or Software Requirements Specification (SRS) (d) User Manual (e) Design Blueprint
35. The professional who acts as a bridge between business users and technical teams is called a _____. (a) Software Tester (b) Project Manager (c) Systems Analyst (d) Database Administrator (e) UX Designer
36. Which of the following is a key property of a good SRS document? (a) It should specify how to implement the system. (b) It should be verbose to cover all details. (c) It should have a "black-box" view, specifying what the system does, not how. (d) It should be written in technical jargon only developers understand. (e) It should avoid documenting non-functional requirements.

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37. What is a major problem an organization might face without an SRS document? (a) The software would be developed too quickly. (b) The system may not be implemented according to customer needs. (c) Developers would have more creative freedom. (d) The project budget would be significantly reduced. (e) Testing would become easier.
38. Which of the following is an example of a functional requirement? (a) The system shall be available 99.9% of the time. (b) The system shall load the homepage within 2 seconds. (c) The system shall allow users to reset their password. (d) All user data shall be encrypted. (e) The system shall be compatible with Chrome and Firefox.
39. Which of the following is an example of a non-functional requirement? (a) The system shall generate monthly sales reports. (b) The system shall calculate employee taxes. (c) The system shall authenticate users via a login screen. (d) The system shall support up to 10,000 concurrent users. (e) The system shall allow administrators to delete user accounts.
40. The "Goals of Implementation" section in an SRS does NOT typically include one of the following (a) Ensuring quality and reliability. (b) Adhering to budget and schedule. (c) Specifying the programming language to be used. (d) Facilitating maintenance and scalability. (e) Ensuring data security and privacy.

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65

70-100 A

60-69 B

40-59 F

0-39 F

70-100 A

65-69 B

60-69 C

50-59 D

40-50 E

0-39 F

2. A



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

COURSE: U-INS212 Management Information System

CREDIT UNIT: 03

TIME: 50 minutes

INSTRUCTIONS: Attempt all Questions and shade the option corresponding to the answer chosen.

1. What is the primary objective of a Management Information System (MIS)?
(a) Replace managerial decision-making (b) Support organizational functions and decision-making (c) Increase operational costs (d) Eliminate communication within organizations (e) Store data only for future reference
2. Which activities are involved in transforming raw data into useful information?
(a) Input, processing, output (b) Storage, retrieval, deletion (c) Analysis, filtering, exporting (d) Capturing, encrypting, deleting (e) None of the above
3. Which component of MIS is responsible for collecting raw data from various sources?
(a) Data Storage (b) Data Processing (c) Data Collection (d) Information Output (e) User Interface
4. Which of the following is NOT a type of MIS? (a) Transaction Processing System (TPS) (b) Decision Support System (DSS) (c) Executive Information System (EIS) (d) Human Resource Management System (HRMS) (e) Customer Relationship Management (CRM)
5. What is the role of Decision Support Systems (DSS)? (a) Process daily transactions (b) Assist in making complex decisions using analytical models (c) Provide top-level executives with summaries only (d) Manage customer interaction exclusively (e) Store employee data only
6. Which of the following is an example of a Customer Relationship Management (CRM) system?
(a) SAP ERP (b) Salesforce Chatter (c) Oracle DBMS (d) Google Drive (e) Microsoft Excel
7. Why are complementary assets important for MIS? (a) They replace the need for technology investments (b) They help maximize the value of information systems (c) They reduce the need for data collection (d) They eliminate the need for organizational change (e) They are optional for effective MIS implementation
8. Which of the following is NOT considered a complementary asset? (a) Organizational culture and leadership (b) Business processes (c) Human capital and skills (d) Software licensing fees (e) Infrastructure and technology
9. Which discipline contributes to the technical approach to studying MIS? (a) Sociology (b) Economics (c) Computer Science (d) Psychology (e) Management
10. The sociotechnical systems perspective emphasizes the importance of:
(a) Only technological features (b) Only social features (c) The best fit between social and technical features (d) Ignoring organizational culture (e) Hardware over software
11. Which component of an information system involves converting raw data into meaningful information?
(a) Data Collection (b) Data Storage (c) Data Processing (d) Information Output (e) User Interface
12. Which of the following is an example of a support activity in the value chain?
(a) Inbound logistics (b) Operations (c) Human resources management (d) Outbound logistics (e) Sales and marketing
13. The process of comparing your business processes against industry standards is called:
(a) Benchmarking (b) Outsourcing (c) Standardization (d) Benchmarking (e) Optimization
14. Which of the following best describes a digital firm?
(a) A company that operates solely offline (b) A firm where digital technologies are central to operations and customer interactions (c) A company that does not use internet technologies (d) A traditional brick-and-mortar business with no online presence (e) A firm that only uses manual processes
15. Which of the following is NOT a benefit of being a digital firm?
(a) Enhanced efficiency (b) Improved decision-making (c) Greater customer engagement (d) Increased operational costs (e) Innovation
16. Which element of a business process involves resources needed to perform tasks?
(a) Inputs (b) Activities (c) Outputs (d) Stakeholders (e) Controls
17. Which level of management is responsible for long-term strategic decisions? (a) Senior management (b) Middle management (c) Operational management (d) Data workers (e) Production workers
18. The physical equipment used in an information system is called: (a) Data (b) Software (c) Hardware (d) Network (e) Cloud service
19. Which of the following is a type of network that is accessible only within a company?
(a) Internet (b) Extranet (c) Intranet (d) Web server (e) Cloud network

20. Which of the following is NOT a component of the IT infrastructure? (a) Hardware (b) Software (c) Data management technology (d) Employee training programs (e) Telecommunications technology
21. E-commerce primarily involves: (a) Internal communication within a company (b) Buying and selling of goods/services over the internet (c) Internal management of business processes (d) Government data management (e) Physical retail store transactions only
22. Which platform is known for enabling employees to contribute and edit content easily? (a) Wikis (b) ERP systems (c) CRM systems (d) Data warehouses (e) E-commerce platforms
23. The term "synergies" in a value web context refers to: (a) Independent operations of firms (b) Sharing outputs as inputs to reduce costs and increase profits (c) Reducing communication among partners (d) Eliminating supply chain relationships (e) None of the above
24. Which of the following best explains "core competencies"? (a) Activities performed by entry-level employees (b) Activities in which a firm is a global leader and excels (c) Activities that are outsourced (d) Tasks that are unimportant for competitive advantage (e) All of the above
25. Which strategy involves targeting a specific segment of the market? (a) Low-cost leadership (b) Product differentiation (c) Focus on market niche (d) Customer and supplier intimacy (e) Broad market approach
26. Which of the following is NOT a typical function of MIS? (a) Supporting operational efficiency (b) Facilitating strategic planning (c) Replacing senior management decision-making (d) Performance monitoring (e) Providing decision-making support
27. Which of the following enables organizations to share information and collaborate across locations? (a) E-mail and instant messaging (b) Business Process Modelling Notation (BPMN) (c) Value stream mapping (d) SWOT analysis (e) All of the above
28. Which role is responsible for translating business problems into information requirements? (a) CIO (b) Data entry clerk (c) System analyst (d) End user (e) HR manager
29. The term "extranets" refers to: (a) Internal company Web sites only accessible by employees (b) Public websites open to everyone (c) Company Web sites accessible to external vendors and suppliers (d) The entire internet infrastructure (e) None of the above
30. Which of the following is a benefit of strengthening customer and supplier intimacy? (a) Increased costs (b) Reduced customer loyalty (c) Improved innovation and efficiency (d) Decreased collaboration (e) Less reliable supply chains
31. Which of these is an example of a primary activity in the value chain? (a) Inbound logistics (b) Operations (c) Human resources management (d) Outbound logistics (e) Sales and marketing
32. Which role is responsible for managing the firm's information technology strategy? (a) Chief Security Officer (CSO) (b) Chief Information Officer (CIO) (c) Chief Privacy Officer (CPO) (d) Chief Financial Officer (CFO) (e) Human Resources Manager
33. Which of the following best describes a "value web"? (a) A single company's internal workflow system (b) A network of interconnected firms that coordinate their processes to respond rapidly to supply and demand changes (c) A traditional supply chain (d) A marketing strategy for online advertising (e) None of the above
34. Which system is primarily used to handle daily transactions like billing? (a) Decision Support System (DSS) (b) Transaction Processing System (TPS) (c) Executive Information System (EIS) (d) Customer Relationship Management (CRM) (e) Business Intelligence (BI) System
35. Which of the following is a key benefit of using MIS for strategic planning? (a) Increased operational costs (b) Better insights into trends and forecasts (c) Eliminating decision-making (d) Reducing data collection efforts (e) None of the above
36. Which component of an information system encompasses data management technology? (a) Hardware (b) Software (c) Data management technology (d) Network (e) User interface
37. Which of the following best explains "mass customization"? (a) Producing identical products for all customers (b) Customizing products using the same production resources as mass production to meet individual customer needs (c) Standardizing all products for efficiency (d) Outsourcing all custom work (e) None of the above
38. What is the main role of a systems analyst? (a) Write code for applications (b) Translate business requirements into information systems (c) Manage hardware infrastructure (d) Oversee data entry tasks (e) Handle marketing strategies
39. The dominant global network that links millions of networks is called: (a) LAN (b) Intranet (c) Internet (d) Extranet (e) VPN
40. The practice of comparing your business processes against industry standards is called: (a) Benchmarking (b) Outsourcing (c) Data mining (d) Standardization (e) Process reengineering



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

COURSE: IFT212 - COMPUTER ARCHITECTURE

CREDIT UNIT: 03

TIME: 50 MINUTES.

INSTRUCTION: ANSWER ALL QUESTIONS

1. introduced the term computer architecture while working at IBM, and the concept was formalized in what year? A. Intel, 1968 B. Microsoft, 1944 C. IBM, 1964 D. John Von Neumann, 1946 E. None of the above ✓
2. Which of the following components carries control signals from the processor to other hardware components? A. Address Bus B. Control Bus C. Data Bus D. Bus E. None of the above ✓
3. is the duration it takes to access a word from memory once a request has been made. A. Cycle Time B. Fetch Time C. Memory Access Time D. Register Delay E. None of the above ✓
4. is a small high-speed memory location within the processor that holds data and instructions currently in use. A. Cache B. Register C. ROM D. Memory Module E. None of the above ✓
5. What does the acronym RISC stand for in computer architecture? A. Reduced Internal Storage Component B. Reduced Instruction Set Computing C. Read Instruction Sequential Code D. Random Instruction Set Computer E. None of the Above ✓
6. is a subsystem that facilitates and sharing of resources between components within a computer or across multiple systems. A. Interface Unit B. Control Unit C. Computer Box D. Computer Bus E. None of the above ✓
7. Which of the following systems is designed to support high-resolution graphics and specialized computing applications? A. Supercomputer B. Server C. Workstation D. Mainframe E. None of the above ✓
8. What is the process of executing jobs in a group rather than individually called? A. Batch Programming B. Time Sharing C. Batch Processing D. Multitasking E. None of the above ✓
9. is a computing technique that divides tasks into smaller sub-tasks to allow simultaneous execution, thereby improving speed. A. Distributed Computing B. Multi-core Processing C. Time Sharing D. Parallel Computing E. None of the above ✓
10. The is a type of ROM that is erasable only by electrical means. A. EPROM B. DRAM C. EEPROM D. SRAM E. None of the above ✓
11. The architecture uses larger instruction sets and multiple addressing modes to execute complex instructions in fewer lines of code. A. RISC B. MISC C. VLIW D. CISC E. None of the above ✓
12. Older PCs typically used processors operating at speeds of about while newer PCs use processors operating at about A. 8-bit, 3.7 MB; 64-bit, 4.66 GB B. 16-bit, 3.9 MB; 64-bit, 3.66 GB C. 32-bit, 2.0 MB; 128-bit, 6.1 GB D. 4-bit, 2.5 MB; 32-bit, 3.5 GB E. None of the above ✓
13. A is a single computer system that contains more than one processing unit sharing memory and other system resources. A. Multi-Computers B. Multiprocessor System C. Microcomputer D. Microprocessor E. None of the above ✓
14. The is a register that keeps track of a program's execution and stores the address of the next instruction to be executed. A. Stack Pointer B. Instruction Register C. Base Register D. Program Counter E. All of the above ✓
15. Who proposed the concept of computer architecture for the EDVAC, and in what year was it completed? A. Blaise Pascal, 1946 B. Charles Babbage, 1837 C. John von Neumann, 1950 D. Jacquard Loom, 1968 E. None of the above ✓
16. What is the full meaning of the acronym POST in computer architecture? A. Power-On Security Test B. Processor Output Self-Test C. Power-On Self Test D. Performance Operating System Test E. None of the above ✓
17. The component ensures that all parts of the computer system are working correctly before the operating system is loaded. A. RAM B. BIOS C. Processor D. GPU E. All of the above ✓
18. Which of the following best describes stored concept of the Von Neumann architecture? A. Programs are hardwired into separate control units B. Data and instructions use separate memory modules C. Each instruction is executed in parallel with the next D. Instructions and data are stored together in the same memory E. None of the above ✓
19. What does an opcode specify in a machine language instruction? A. The amount of memory needed B. The operation to be performed C. The address of the data D. The data to be used E. None of the above ✓
20. What is considered the core component of an operating system that manages system resources? A. Kernel B. Scheduler C. Loader D. Compiler E. None of the above ✓

21. In the context of machine instructions, what is the primary role of operands? A. Store the instruction B. Specify the operation ☒ C. Provide data for the operation D. Initiate program execution ☐ E. None of the above
22. Which of the following best differentiates computer architecture from computer model? ☒ A. Architecture focuses on physical design; model focuses on logical design B. Architecture is an abstract platf. model is a specific implementation C. Architecture is hardware, model is software D. Architecture and model are synonymous terms ☐ E. All of the above
23. What is the primary purpose of the instruction register in a CPU? A. Stores final results of operations B. Keeps track of memory size ☒ C. Holds the current instruction being executed D. Stores previous user inputs E. All of the above
24. In which generation of computers were high-level programming languages like FORTRAN and COBOL introduced? A. First generation ☒ B. Second generation C. Third generation D. Fourth generation E. None of the above
25. Which of the following best describes how the control unit manages instruction execution in modern CPUs with pipelining? A. It compresses instruction cycles to reduce memory usage B. It performs arithmetic and logic functions in parallel ☒ C. It supervises simultaneous stages of multiple instructions to increase throughput D. It fetches and executes one instruction at a time in strict sequence E. None of the above
26. Which of the following best describes addressing modes in instruction execution? A. They refer to the physical layout of memory cells. B. They are responsible for assigning memory addresses during boot time. ☒ C. They define how an instruction determines the location of its operands. D. They determine how the CPU controls data buses. E. None of the above
27. What is a significant challenge in relation to Von Neumann architecture known as the 'Von Neumann bottleneck'? A. Limited instruction decoding per clock cycle B. Simultaneous access to memory by CPU and GPU. C. Inability of the CPU to execute conditional jumps. ☒ D. Shared bus for data and instructions slows down throughput. E. None of the above
28. The primary advantage of using registers in a CPU is? ☒ A. Faster access speed compared to other memory types B. Higher capacity than RAM C. Greater non-volatility D. Ability to store programs permanently E. All of the above
29. Which error detection technique appends a single bit to a binary sequence? A. Redundancy Code ☒ B. Parity Bit C. Checksum D. CRC E. Kermit Code
30. The acronym ALU stands for: A. Advanced Level Utility B. Arithmetic Linear Unit ☒ C. Arithmetic Logic Unit D. Automated Logic Unit E. Advanced loop unit
31. Which of these components directly controls instruction execution timing? A. Bus Interface B. Memory Unit C. ALU ☒ D. Control Unit E. Registers
32. A computer bus is responsible for? A. Powering the CPU B. Connecting input devices to output ☒ C. Transferring data between components D. Interpreting machine code E. All of the above
33. A 'nibble' consists of? A. 2 bits ☒ B. 4 bits C. 8 bits D. 16 bits E. 32 bits
34. Which of the following is not a valid performance metric in computer systems? A. MIPS B. FLOPS ☒ C. RPM D. CPI E. None of the above
35. The main difference between static and dynamic RAM is? ☒ A. Speed of access B. Whether they are volatile ☒ C. Use of transistors vs capacitors D. Use in ROM vs RAM modules E. All of the above
36. The system clock in a CPU is essential for? A. Permanent data storage B. Instruction decoding ☒ C. Synchronizing operations D. Managing I/O requests E. All of the Above
37. The instruction register (IR) stores? A. The final output B. The next memory address ☒ C. The currently executing instruction D. ALU output E. All of the above
38. Which of the following best defines instruction pipelining? ☒ A. Parallel execution of data B. Storing multiple instructions in a queue ☒ C. Overlapping instruction phases to improve performance D. Using dual processors E. None of the above
39. The main function of a decoder in the instruction cycle is to: ☒ A. Translate machine code into micro-operations B. Fetch data from RAM C. Execute instructions D. Signal errors E. Overlap instructions.
40. Which of the following does not affect CPU performance? A. Clock speed B. Number of registers ☒ C. Brand name of the manufacturer D. Instruction set complexity E. None of the above



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS
COURSE: INS204 – SYSTEM ANALYSIS AND DESIGN
CREDIT UNIT: 03

TIME: 1 hour

INSTRUCTION: Answer all questions

1. Which of the following best defines a system? A. A random collection of components
B. A group of independent entities C. A set of interrelated components working towards a goal ✓
D. A file structure used for programming E. A user interface
2. Which of these is NOT a common type of system? A. Manual B. Automated C. Hybrid
D. Transparent E. Open ✓
3. Which phase of the SDLC involves determining the needs of the end user? A. Design
B. Implementation C. Requirements analysis D. Testing E. Deployment ✓
4. A major objective of system approach is to: A. Eliminate all human roles
B. Focus only on technical requirements C. View problems holistically ✓
D. Reduce the number of components E. Avoid feedback
5. A non-functional requirement includes: A. User must log in with a password
L B. System must support login authentication C. System should respond within 2 seconds
D. Sales data must be stored E. Database should contain 10 tables
6. Which of the following is NOT part of system requirements? A. Functional requirements B. Legal constraints
C. Physical location of the user D. Hardware interface E. Organizational structure ✓
7. A level-0 DFD is also known as: A. Primitive DFD B. Abstract DFD C. Context diagram D. Flowchart
E. Control diagram L
8. External evaluation might involve: A. Internal managers B. Programmers C. System users only
D. Independent auditors E. Database designers L
9. Which of the following is NOT a benefit typically evaluated? A. Time saving B. User satisfaction
C. Developer interest D. Cost reduction E. Flexibility ✓
10. One of the first steps in system implementation is. A. Launching the marketing campaign B. Creating test data
C. Developing a plan D. Interviewing users again E. Deleting old files L
11. A deterministic system is defined by: A. Multiple simultaneous outputs for a single input
B. High dependency on environmental feedback C. Predictable behavior under specified inputs
D. Adaptive real-time learning modules E. Self-regulating feedback loops
12. Which SDLC model is best suited for systems with rapidly changing requirements? A. Waterfall
B. Spiral C. V-model D. Big Bang E. Agile L
13. A technical requirement can be differentiated from a system requirement by its: A. Broad organizational focus
B. Abstract goal orientation C. Precise implementation details D. Subjectivity to stakeholder interpretation
E. Legal and regulatory conformance ✓
14. Which phase of SDLC most critically depends on a complete understanding of non-functional requirements? A. Design
B. Maintenance C. Analysis D. Testing E. Deployment L
15. A feasibility study that assesses whether the organization has the capacity to support a new system is: A. Technical B. Operational
C. Economic D. Legal E. Schedule ✓
16. Which of the following cost comparisons best helps in determining Return on Investment (ROI) for a proposed system? A. Development vs. maintenance cost
B. Current system operation cost vs. proposed system operation cost ✓

- new system operation cost C. Vendor vs. in-house development cost D. Licensing vs. subscription cost E. UI design vs. backend optimization cost
17. Feasibility analysis often omits in early-stage decision-making.
A. Legal feasibility ~~B. Emotional resistance from users~~ C. Budgetary constraints ✓
D. Timelines for development E. Technological compatibility
 18. Which feasibility type is best evaluated using Monte Carlo simulation or probabilistic forecasting?
A. Operational B. Schedule C. Technical D. Legal ~~E. Economic~~ ✓
 19. Which DFD level represents the most granular view of system processes?
A. Level-0 B. Level-1 C. Level-2 D. Context diagram E. Entity-Relationship Diagram ✓
 20. Which of the following is a direct consequence of incorrect analysis specifications?
~~A. Inconsistent documentation~~ B. Accurate test data C. Optimized file conversion ✓
D. Accelerated system evaluation E. Enhanced network throughput
 21. The primary role of a feasibility report is to:
A. Describe coding architecture B. Provide user manuals C. Justify the system investment ✓
D. Design ER diagrams E. Test data accuracy
 22. Which fact-finding technique yields the least biased data?
A. Interviews B. Questionnaires C. Observation ~~D. Prototyping~~ E. Document analysis ✓ 38
 23. Which type of requirement is typically identified after user shadowing? A. Functional
~~B. Operational~~ C. Technical ~~D. Non-functional~~ E. Regulatory ✓
 24. Which of the following is a major limitation of document review in fact-finding?
A. High cost B. Subjectivity C. Time inefficiency ~~D. Inaccuracy due to outdated information~~
E. Limited scalability
 25. In requirements engineering, traceability primarily helps in: A. User feedback analysis
B. Defining system boundaries C. Ensuring complete documentation ~~D. Linking needs to deliverables~~ ✓
E. Training stakeholders
 26. What is the most significant challenge in producing technical documentation for system design?
A. Writing in natural language ~~B. Keeping documentation up-to-date~~ C. Visualizing user interfaces ✓
D. Translating pseudocode E. Explaining data types
 27. Which system specification task is most dependent on prior feasibility findings?
A. Interface layout B. Logic flow design ~~C. Process modeling~~ ~~D. Hardware platform choice~~ ✓
E. Access control configuration
 28. Which diagram is least useful for specifying logical process design? A. Flowchart B. Pseudocode
C. state diagram ~~D. ER Diagram~~ E. DFD ✓
 29. Which aspect of documentation is most critical for system maintainers? A. Marketing documentation
B. User Interface wireframes ~~C. System architecture blueprints~~ D. Help manuals E. Deployment checklists ✓
 30. What ensures consistency between design documentation and implementation? A. Regular code refactoring
B. Version control of diagrams ~~C. Continuous integration~~ ~~D. Traceability matrix~~ ✓
E. Data dictionary
 31. In system implementation, which factor is most crucial when planning parallel runs? A. Hardware availability
B. User resistance level C. Data redundancy D. System load forecasting E. Cost of dual operation ✓
 32. What is the most strategic function of the Education Coordinating Committee during system implementation? A. Approving budget allocations
B. Configuring hardware modules ✓
~~C. Managing user training consistency~~ D. Writing user manuals E. Designing the UI
 33. Which of the following best exemplifies a white-box test case during system test data preparation?
~~A. Validating UI flow for errors~~ B. Stress-testing a login form ~~C. Inputting invalid data to check crash behavior~~ ✓
D. Verifying loop conditions and branches E. Analyzing black swan scenarios

34. The key reason for post-implementation reviews is: A. Budget verification B. Performance benchmarking C. Stakeholder negotiation D. Regulatory auditing E. Data center optimization
35. A poor system test data plan most often leads to A. Project scope creep B. User interface inconsistency C. Functional redundancies D. Latent system faults E. Inadequate documentation
36. Which of the following file conversion challenges is most likely to impact data integrity?
☒ A. Index corruption B. Batch job failure C. Operating system mismatch
 D. Memory leak E. Multithreaded conflict
37. What is the main responsibility of the data control staff during system installation?
 A. Defining new record structures B. Performing test case analysis ☒ C. Monitoring data movement and validity D. Managing file permissions E. Writing software patches
38. During installation, data preparation staff are expected to: A. Program data entry screens
 B. Configure network protocols ☒ C. Reformat and validate data sets
 D. Review system logs E. Conduct user evaluations
39. The greatest risk of direct installation strategy is: A. Delayed stakeholder approval
 B. High testing overhead ☒ C. Lack of fallback system D. Redundant staffing E. Legal compliance
40. Which approach most effectively mitigates data migration risks during installation? A. Load balancing
☒ B. Parallel systems comparison C. Server caching D. Disk mirroring E. Incremental rollout
41. Which of the following would be considered an intangible benefit in system evaluation?
 A. Reduced server maintenance cost B. Decreased processing time ☒ C. Improved customer satisfaction D. Fewer full-time employees E. Lower electricity bills
42. System evaluation that focuses on regulatory compliance falls under: A. Operational analysis
☒ B. Cost-benefit assessment ☒ C. External audit requirement D. Technical validation
 E. Strategic review
43. Which evaluation method best assesses system performance under variable load conditions?
☒ A. Focus group feedback B. Regression testing ☒ C. Stress testing D. Cost-to-benefit ratio
 E. Risk assessment matrix
44. What is the main purpose of post-implementation recommendations? A. Reduce developer workload
 B. Suggest training schedules ☒ C. Ensure continuous improvement D. Justify project delays
 E. Identify regulatory violations
45. Which system evaluation metric is most susceptible to subjective bias? A. Net present value
☒ B. Usability score C. Throughput rate D. Downtime duration E. Defect density
46. A system that fails to meet an external requirement will likely face: ☒ A. Cost escalation B. Increased user training
☒ C. Legal consequences D. Storage bottlenecks E. Redundant scripting
47. Which scenario represents a negative net benefit in cost-benefit analysis? ☒ A. Training costs outweigh efficiency gains
 B. Revenue increase after system adoption C. Maintenance costs decrease over time D. ROI greater than 1 E. Break-even in under a year
48. Which of the following best supports quantitative evaluation of systems success? A. Employee morale surveys
 B. Incident report reviews ☒ C. Metrics on processing delays D. Client testimonials
 E. Stakeholder interviews
49. A benefit realization plan is used in evaluation to: A. Document project risks ☒ B. Translate outputs into measurable outcomes
 C. Allocate task resources D. Align team roles with deliverables
 E. Approve technical constraints
50. A key limitation of cost-benefit analysis in systems evaluation is: A. High level of automation
☒ B. Lack of documentation ☒ C. Inability to capture long-term impacts accurately D. Ease of duplication
 E. Requirement for software modeling